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The Effects of Feed Frame Parameters and Turret Speed on Mini-Tablet Compression

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ABSTRACT

Die filling is a critical process step during tablet production as it defines the tablet weight. Achieving die fill consistency during production of mini-tablets, tablets with diameters ≤6 mm, is considerably more challenging. Although die filling in rotary presses had been studied in relation to the feed paddle design, paddle speed, and turret speed, it is unclear how these process variables could impact mini-tablet production and product properties. In this study, 1.8 and 3 mm mini-tablets were prepared using a rotary press with multiple-tip tooling using different process configurations. Mini-tablet weight variation within and across compaction cycles were determined using data from compression roller displacement and mini-tablet weight. Higher die fill densities were achieved with a flat feed wheel paddle and high paddle speed. This was attributed to better granule fluidization in the feed frame, which also increased the intercycle weight variation and reduced tensile strength. The turret speed did not impact mini-tablet properties significantly. Granule overlubrication in the feed frame potentially reduced mini-tablet tensile strength during compaction. The number of paddle passes in the die fill region was correlated to mini-tablet die fill performance. Findings from this study could provide better insights into the relationship between process variables and mini-tablet product quality.

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Introduction

Tablets are usually manufactured on high-speed rotary tablet presses where powder blends in feed frames are fed into dies for compaction. Reproducibility of the die filling step determines tablet weight variability. Poor die fills result in inhomogeneous fill densities, which could affect other critical mechanical attributes of the tablets produced, such as tensile strength. Consistent die fill is therefore one of the most critical prerequisites to control tablet quality and sometimes may be the rate-limiting factor during rotary tablet production.[1,2] It is conceivable that the task of achieving consistent die fill during production of mini-tablets, tablets with diameters smaller than 6 mm, is much more daunting than conventional-sized tablets. Clearly, both the narrowness of the die orifice and the low tablet weight are impediments to consistent mini-tablet quality. Despite these challenges, there is strong interest in the rational development of mini-tablets to be manufactured commercially as they offer distinct advantages over conventional-size tablets, such as greater ease of swallowability, suitability for the young, dosing flexibility, and potential for individualized drug therapy.[3-5]

Each compaction cycle on a rotary tablet press comprises die filling, compression, and ejection. The act of die filling involves several mechanisms working concurrently or in tandem: gravity feeding where feed enters the die orifice under the effect of gravity; forced feeding by rotating wheels in the feed frame pushing feed toward the die orifice; and suction fill effected by partial vacuum created by the downward displacement of the lower punch, which draws feed into the die orifice.[6,7] Feed frames are ubiquitous features of rotary tablet presses, and each unit serves to aerate the feed material and deliver it to the slit at the bottom of the feed frame for deposition into moving dies.

Depending on the tablet press model, feed frames typically contain several coplanar chambers in which the feed wheel and metering wheel rotate continuously. The “arms” or paddles of the wheels can come in different shapes and profiles.[7,8] Several studies have evaluated powder phenomena inside the feed frames and its effect on powder and tablet properties. Through computational simulations and experimental setups, it has been shown that increasing the wheel speed could enable higher die fill densities,

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lower tablet weight variation, and improve flow properties of lubricated feed material.[1,9-11] However, prolonged blending inside the feed frame could be detrimental by increasing lubricant-related hydrophobicity, reduced dissolubility, and lower tablet strength.[12-14] The wheels in the feed frame could also apply shear forces to the feed granules, causing unintended attrition and changes in size distribution and bulk densities.[2,15] The effects of wheel design (number of paddles, paddle thickness, and hub diameter) on powder residence time inside the feed frame and tablet weight have also been investigated.[9,11] More recently, inline near-infrared spectroscopy has been used to study powder mixing dynamics in the feed frame, and it showed changes in blend potency and powder densities within the feed frame.[16,17]

Studies performed with shoe-die models showed that die diameter and interactions between feed particles entering the die and air exiting the die influenced die fill dynamics.[18,19] Gravity fill into a die cavity is primarily governed by nose flow, bulk flow, and intermittent flow.[19-21] As the feed shoe traverses across the die opening, the initial acceleration of the feed shoe coupled with frictional interactions between the powder and die platform causes the powder to adopt a nose-shaped profile.[7,21] When the tip of the "nose" initially moves across the die opening, particles cascade down the slope of the nose into the die, exhibiting nose flow. Air gets displaced out from the die cavity as powder enters into it. As the feed shoe continues to move across the die opening, it eventually covers the die opening completely and powder can only enter the die cavity through detachment from the bottom of the bulk powder base in the shoe over the die orifice. If powder flow behavior into the cavity is smooth, it is termed as bulk flow. In cases where powder detachment occurs randomly into the die, the flow is known as intermittent flow. The concept of critical velocity or maximum feed shoe speed at which the die orifice could be filled completely with one pass of the feed shoe was used as an alternate measure of powder flowability.[8,20,22] Correlations between powder flow properties and die fill variation have also been investigated.[7,23-26]

The interplay of gravity fill, forced feeding, and suction effect makes die filling in rotary tablet presses a rather complex process. Mini-tablet production usually requires multiple-tip tooling, which adds complexity to the die fill process as weight variations can occur within and across compaction cycles. Although the feed frame plays an important role in die fill on rotary tablet presses, much of the available literature examined die fill in other contexts and did not take into account the complex features of an actual production feed frame.[27] Moreover, the findings from the various studies are not specific to mini-tablet production.

Main process variables that could affect die filling in rotary tablet presses include the feed wheel paddle design, feed wheel paddle speed, and turret speed. However, the effects of these process variables on mini-tablet physical properties are still not well understood. The aim of this work was to investigate the influence of these variables on mini-tablet weight, weight variation, and tensile strength. Pharmaceutical granules were compacted into 1.8 mm and 3 mm mini-tablets on a rotary tablet press at different turret and feed wheel speeds. Different wheel paddle profiles were used to alter the forced feeding effect.

Materials and Methods

Materials

Lactose (GranuLac 200 M, Meggle, Germany) and polyvinylpyrrolidone (PVP K25; Kollidon 25, BASF, Germany) formed the powder mass for granulation. Purified water was used as the granulating liquid. Magnesium stearate (MgSt; M125, Productos Metalast, Spain) was used as the lubricant.

Granule Preparation

High-Shear Granulation

Lactose and PVP K25 (95:5) were dry blended in a 10 L high-shear mixer (PMA-1, GEA Niro Pharma Systems, UK) for 3 min at an impeller speed of 500 rpm. Granulating liquid, which constituted 10%, w/w of the total dry powder weight, was sprayed via a pressure pot onto the powder mass over 3 min. Wet massing continued for an additional 1 min with an impeller speed of 650 rpm. The wet granules were delumped through a conical screen mill (Comil® 197S, Idex-Quadro Engineering, Canada) fitted with a 6450 μm square aperture sieve and a square impeller rotating at 1239 rpm. Granules were dried using a fluid bed processor (STREA-1, GEA-Aeromatic, Switzerland) at an inlet air temperature of 60°C, and drying stopped when an exhaust temperature of 40°C was attained. The dried granules were regranulated using the conical screen mill fitted with a 610 μm round aperture screen and a round impeller rotating at 2039 rpm.

Granule Lubrication

The granules were blended with MgSt in a double cone blender rotated at 10 rpm for 3 min before flow characterization and tableting. The minimal amount of MgSt required for lubrication and successful tableting without sticking issues was determined for each mini-tablet diameter to avoid risk of overlubrication. For 1.8 mm mini-tablets, 1%, w/w MgSt was found to be sufficient. A higher level of 2%, w/w MgSt was necessary for 3 mm mini-tablets to ensure low ejection forces and no sticking. Although higher die-wall frictional forces were expected with the 1.8 mm mini-tablets, the greater fill depth required for the 3 mm mini-tablets entailed a longer ejection distance that warranted the higher lubricant requirement.

Granule Particle Size Characterization

Granule size and size distribution were determined using a laser diffractometer (LS230, Beckman Coulter) fitted with the dry powder module. Granules were delivered at a controlled rate to attain an obscuration level of about 10%. Particle size parameters corresponding to the 10th, 50th, and 90th percentiles (D_{10}, D_{50}, and D_{90}) were determined from the cumulative undersize curve generated. Span, an indicator of size distribution, was calculated using Equation 1. Results were averaged from triplicate readings.

$$Span = \frac{D_{90} - D_{10}}{D_{50}} \quad (1)$$

The particle size and size distribution data were as follows, D_{10}: 78.3 ± 9.0 μm, D_{50}: 216.8 ± 13.0 μm, D_{90}: 386.4 ± 11.4 μm, and Span: 1.42.

Granule Flow Characterization

Determination of Angle of Repose

Angle of repose (AoR) of granules was determined using the fixed base method. Approximately 100 *g* of granule sample was dispensed through a funnel onto a circular base plate (Copley Scientific, UK) of 2r diameter to form a conical heap, and its height, h, measured. The AoR was calculated using Equation 2. Results were averaged from triplicated experiments.

$$AoR = \tan^{-1}\left(\frac{h}{r}\right) \quad (2)$$

Determination of Tapped and Bulk Densities

Granule samples were gently sifted through a 1 mm aperture sieve into a preweighed cylinder cut exactly to 100 cm^3. Excess particles were carefully removed from the top of the cylinder, adhering powder dusted off and weighed. The filled cylinder was tapped (J. Engelsmann AG, Germany) at 250 drops/min to a constant tapped volume. The experiments were triplicated. Bulk (ρ_{bulk}) and tapped (ρ_{tapped}) densities were calculated using Equations 3 and 4, respectively, and used to determine the Hausner ratio using Equation 5.

$$\rho_{bulk} = \frac{Sample\ mass}{100\ mL} \quad (3)$$

$$\rho_{tapped} = \frac{Sample\ mass}{Tapped\ volume} \quad (4)$$

$$HR = \frac{\rho_{tapped}}{\rho_{bulk}} \quad (5)$$

Powder Rheometry

Basic flow energy (BFE) and specific energy (SE) of the granules were determined using a powder rheometer (FT 4, Freeman Technology, UK). A twisted blade moved in a series of downward clockwise and upward anticlockwise rotations through the granule sample in a glass cylinder while measuring the energy required for the granule flow. BFE was the amount of energy measured during the blade's downward traverse under conditions of high stress flow. SE was determined during the upward traverse, where the particles flowed in a gentler manner. Results were averaged from triplicate measurements.

Avalanche Flow

Approximately 100 cm^3 of granule sample was rotated at 0.6 rpm in a drum with transparent sides to determine its avalanche flow behavior (Revolution® Powder Analyzer, Mercury Scientific). The drum was first rotated for 60 s to aerate the powder and establish a reproducible initial packing state. A digital camera captured images of avalanches and powder behavior for a total of 150 powder avalanches. The avalanche angle (angle of powder inclination before avalanche), avalanche time (duration between consecutive avalanche events), and avalanche energy (amount of energy released during the avalanche event) were determined by the analysis software and average values calculated from triplicate readings.

Mini-Tablet Production

Mini-Tablet Compaction on Rotary Tablet Press

Mini-tablets of 1.8 mm and 3 mm diameters were compacted from lubricated granules on a rotary tablet press (R190FT, GEA, Belgium) installed with multiple-tip, standard concave tooling (1.8 mm punches × 9 tips per die, 3 mm punches × 8 tips per die; Natoli Engineering). To facilitate accurate sampling and collection of mini-tablets during production, only one set of tooling was installed on the press while the remaining compression stations were fitted with blanks. The tablet press was fitted with a fill cam of 8 mm depth. Fill depths were set at 2.4 and 4.5 mm (1.5 × tablet diameter) for the 1.8 mm and 3 mm mini-tablets, respectively. Compression pressure applied per punch tip was 100 MPa.

Two sets of feed and metering wheels were used for the experimental runs: one with a paddle profile of 90° inclination that was supplied with the tablet press and a custom-made set (Natoli Engineering) with a paddle profile of 45° inclination (Fig. 1a). Besides the paddle profiles, the two sets of wheels were otherwise similar in terms of diameter, number of paddles, and paddle thickness.

Full factorial experimental designs for both 1.8 mm and 3 mm mini-tablets at different process configurations were performed: feed wheel paddle profile (flat: 90° and angled: 45°), feed wheel speed (5, 11, and 17 rpm), and turret speed (25, 35, and 45 rpm). Speed of the metering wheel must be higher than that of the feed wheel to avoid excessive compression of material inside the feed frame.[10] As such, the metering wheel was set to rotate at twice the feed wheel speed for each experimental run.

Determination of Number of Paddle Passes Across Die in Die Fill Region

Figure 1b shows a schematic of the turret, feed wheel, and metering wheel on the rotary tablet press. A two-chamber feed frame containing the feed wheel in the first chamber and the metering wheel in the second chamber was used for this study. The wheels were motor-driven and capable of rotating at different speeds independently of each other. Operational details from the rotation of the turret, feed wheel, and metering wheel are indicated in the schematic. Granules enter the die under the influence of the

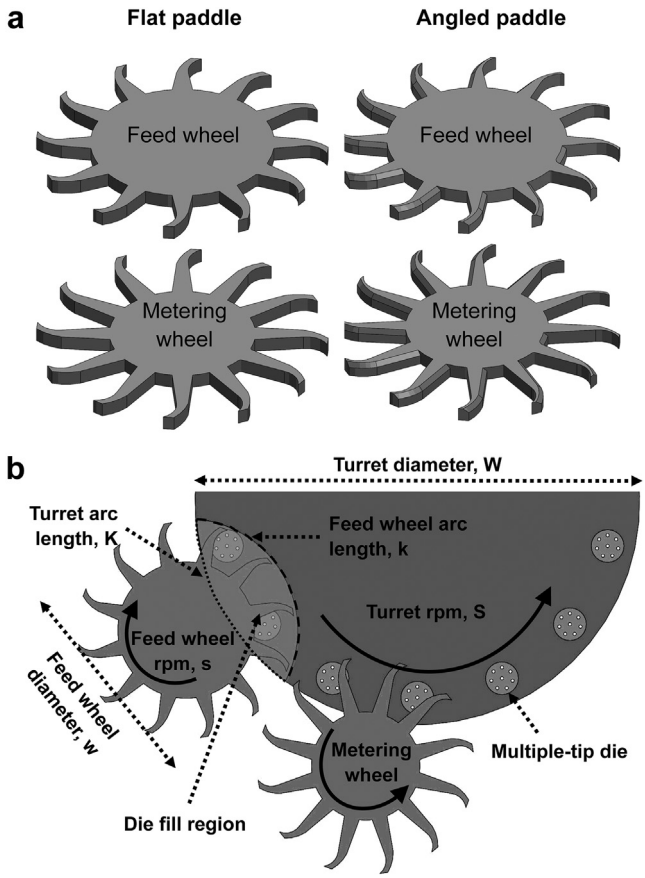


Figure 1. (a) Schematic showing the bottom (reverse) side of the feed wheels and metering wheels used in this study. (b) Schematic of turret, feed wheel, and metering wheel on the rotary press.

feed wheel, and suction is brought about by the rapid descent of the lower punch at the fill cam. A weight adjustment mechanism then raises the lower punch position to the desired fill depth and excess granules are recovered into the feed frame by the scraping action of the metering wheel. In this process, the die passes under a number of feed wheel paddles (N_p) in the die fill region. It was of interest to determine N_p as it represented the relative amount of granules delivered to the die during the filling phase; higher N_p values indicate that more fresh granules were delivered to the die fill region. By taking geometrical measurements of the turret and feed wheel, it is theoretically possible to solve for N_p for a given set of process variables (turret and feed wheel rotational speeds) using Equation 6. Table 1 shows the N_p values for different combinations of the paddle and turret speeds.

$$N_p = \frac{N(kSW + Ksw)}{\pi SWw} \tag{6}$$

where $N = 12$, the number of paddles on feed wheel, k = 200 mm, arc length of feed wheel on turret, K = 175 mm, arc length of turret on feed wheel, w = 200 mm, feed wheel diameter, W = 415 mm, turret diameter, s is the feed wheel rotation speed, and S is the turret rotational speed.

Sampling Protocol for Mini-Tablet Collection

The sampling protocol used for mini-tablet collection has been reported previously[28] and described briefly in this section. For each experimental run, after an initial 100 compaction cycles to condition and stabilize the compaction process, consecutive sampling of mini-tablets was carried out at cycles 101-105, 301-305, and 501-505. Mini-tablets were also collected once every 40 compaction cycles from cycle 120 to 480. Sampling was thus performed for 25 compaction cycles of a total 505 cycles. Sampled mini-tablets were stored for at least 24 h under controlled ambient temperature and relative humidity conditions (25°C, 50% RH) before characterization.

Mini-Tablet Characterization

Intercycle Weight Variation

The rotary press used for this study has a "floating" compression roller held down by a piston that allows the roller to move up vertically by an air compensator mechanism. Unlike rotary presses with a fixed roller where the compression force applied is determined by the tablet weight, the floating roller can move up when the compression force is beyond a set value. This allows control of the peak compression force applied regardless of the tablet weight. A linear variable displacement transducer sensor measures displacement of the roller accurately.

Table 1
N_p Values for Different Process Configurations

Paddle Speed (s, rpm)	Turret Speed (S, rpm)	N_p
5	25	4.14
11	25	4.53
17	25	4.91
5	35	4.05
11	35	4.33
17	35	4.60
5	45	4.00
11	45	4.21
17	45	4.43

Displacement values of the compression roller recorded after the conditioning phase ($n = 405$) were used to monitor mini-tablet weight variation across the compaction cycles. SD of the displacement values was used in Equation 7 to calculate coefficient of variation (COV) and inter-COV to describe mini-tablet weight variation across the 405 compaction cycles.

$$Inter - COV_M = \frac{SD\ of\ roller\ displacement\ across\ cycles}{Mean\ roller\ displacement\ across\ cycles} \times 100\,\% \tag{7}$$

where M is the mini-tablet diameter in mm.

Mini-tablets from each sampled cycle were weighed individually. The mean and SD of mini-tablet weights across the 25 sampled compaction cycles were determined and used in Equation 8 to calculate inter-COV_{Ref}, which was compared against inter-COV.

$$Inter - COV_{Ref} = \frac{SD\ of\ mini - tablet\ weight\ across\ cycles}{Mean\ mini - tablet\ weight\ across\ cycles} \times 100\,\% \tag{8}$$

Intracycle Weight Variation

For each sampled cycle, the mean and SD of mini-tablet weight were determined and used in Equation 9 to obtain intra-COV. Average intra-COV from the 25 sampled compaction cycles was reported.

$$Intra - COV_M = \frac{SD\ of\ mini - tablet\ weight\ within\ sampled\ cycle}{Mean\ mini - tablet\ weight\ within\ sampled\ cycle} \times 100\,\% \tag{9}$$

where M is the mini-tablet diameter in mm.

Mini-Tablet Tensile Strength

Three randomly selected mini-tablets from each sampled compaction cycle were used for breaking force determination with a tensile tester (EZ Test, Shimadzu, Japan) fitted with a 100 N load cell. Mini-tablet thickness was measured with a micrometer screw gauge. Tensile strength of each mini-tablet was calculated using Equation 10, and the average value was determined across all the sampled compaction cycles.

$$Tensile\ strength = \frac{2F}{\pi MT} \tag{10}$$

where F is the breaking force, M and T are mini-tablet diameter and thickness in mm, respectively.

Statistical Analyses

Independent samples t-test, ANOVA, and post hoc tests were conducted at the 95% CI (IBM SPSS Statistics). The same software was used for bivariate linear regression modeling between inter-COV and inter-COV_{Ref}. Partial least squares regression was used for multivariate analysis to evaluate the relative importance of process factors on mini-tablet properties (The Unscrambler® X v.10.3, CAMO Software, Norway). Data scaling and cross validation were performed before multivariate regression analyses. Regression models were evaluated based on their goodness-of-fit (R^2) and root mean square error values.

Table 2
Flow Parameters of Nonlubricated and Lubricated Granules

Lubrication (%, W/W MgSt)	AoR (°)	HR	AvA (°)	AvE (kJ/Kg)	AvT (s)	BFE (mJ)	SE (mJ/g)
0	28.6 (0.6)	1.12 (0)	41.5 (1.0)	10.71 (1.08)	2.40 (0.17)	2313.9 (105.4)	5.84 (0.12)
1	28.7 (0.1)	1.13 (0.02)	38.4 (0.2)	8.54 (0.68)	2.00 (0.10)	963.3 (52.4)	3.18 (0.10)
2	28.8 (0.4)	1.18 (0.01)	39.9 (0.7)	10.36 (1.89)	2.07 (0.12)	1071.0 (44.7)	3.53 (0.19)

Parenthesized values indicate the respective SDs.

Results

Physical Characteristics of Granules

Table 2 shows the granule flow parameters. With the exception of AoR and avalanche energy, lubricated granules generally showed improved flow as evident by the Hausner ratio, avalanche angle, avalanche time, and BFE and SE values, with the most obvious change observed for BFE. However, no significant differences in flow parameters were seen between granules lubricated with 1 or 2%, w/w MgSt ($p > 0.05$ for all parameters). Flow properties of the two lubricated granule batches were thus generally similar and unlikely to confound die fill performance during mini-tablet production.

Effects of Process Parameters on Mini-Tablet Weight

None of the mini-tablets produced exhibited mechanical failures such as capping or lamination. Average weight of 1.8 mm mini-tablets was between 4.7 and 5.1 mg, whereas 3 mm mini-tablets weighed between 21 and 23 mg (Figs. 2a and 3a). For both mini-tablet diameters, there was little apparent difference in weight when using either of the paddle profiles ($p > 0.05$). Increased paddle speed raised mini-tablet weight slightly, for both diameters ($p < 0.05$). Although mini-tablet weight appeared to decrease slightly with an increase in the turret speed, especially for process configurations with the flat paddle, the effect was not significant ($p > 0.05$).

Effects of Process Parameters on Mini-Tablet Tensile Strength

Tensile strength of 1.8 mm mini-tablets ranged between 0.6 and 0.9 MPa, whereas that of 3 mm mini-tablets was between 1 and 1.2 MPa (Figs. 2b and 3b). The paddle angle did not have an apparent effect on 1.8 mm mini-tablet tensile strength ($p > 0.05$). In contrast, 3 mm mini-tablets produced with the angled paddle had higher tensile strength, particularly at lower paddle speeds ($p < 0.05$). For both mini-tablet diameters, tensile strength decreased with increased paddle speed ($p < 0.05$), whereas the turret speed did not appear to have a clear effect on tensile strength ($p > 0.05$).

It was worth noting that mini-tablet tensile strength was not always consistent during production. Figure 4 shows the average mini-tablet tensile strength across compaction cycles for both 1.8 mm and 3 mm mini-tablets produced at the same turret speed of 45 rpm but different paddle profile and speed configurations. For mini-tablets sampled during compaction cycles 101-105, 301-305, and 501-505, the average tensile strength from an aggregate of 5 cycles was reported for each group. Although the tensile strength of 1.8 mm mini-tablets remained relatively constant (Figs. 4a(i) and 4a(ii)), a slight decrease in 3 mm mini-tablet tensile strength was observed (Figs. 4b(i) and 4b(ii)). Similar observations were made at other turret speeds.

Effects of Process Parameters on Intercycle Mini-Tablet Weight Variation

Figure 5a shows the relationships between inter-$COV_{1.8}$ and inter-COV_3 with inter-COV_{Ref} (R^2: 0.59-0.61, $p < 0.01$ in both cases). A possible reason for this could be due to the large difference in sampling size: inter-COV_{Ref} considered only 25 sampled cycles, whereas inter-$COV_{1.8}$ and inter-COV_3 accounted for 405 compaction cycles each. Variation in mini-tablet weight captured by the compression roller displacement might not be captured entirely by that determined from sampled mini-tablets. Nonetheless, coefficients of determination were statistically significant, and the

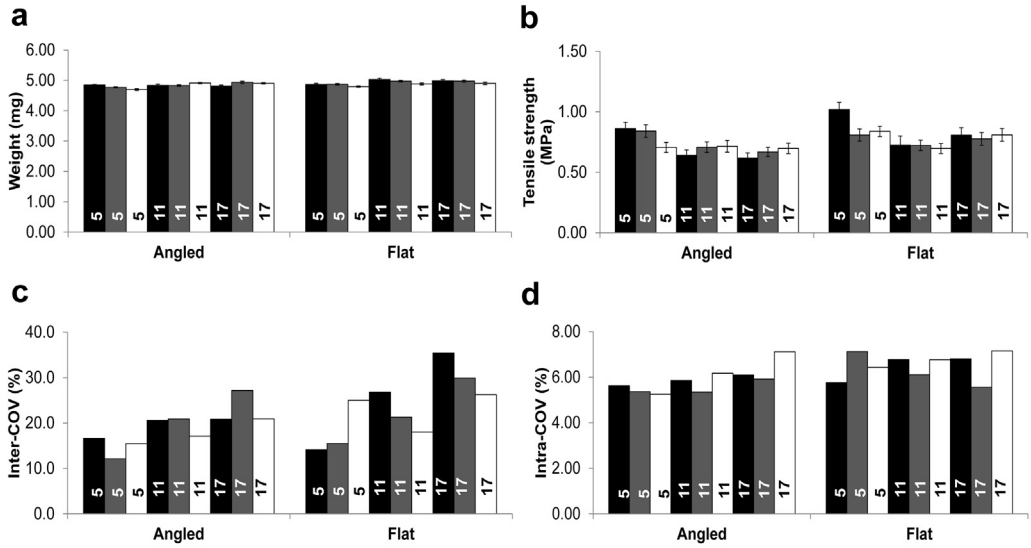


Figure 2. Effects of the paddle angle, paddle speed, and turret speed on (a) average weight, (b) tensile strength, (c) inter-COV, and (d) intra-COV for 1.8 mm mini-tablets. Turret speed: (■) 25 rpm, (▒) 35 rpm, and (□) 45 rpm. The numbers annotated within the bar graphs indicate the paddle speed in rpm. Error bars indicate CIs.

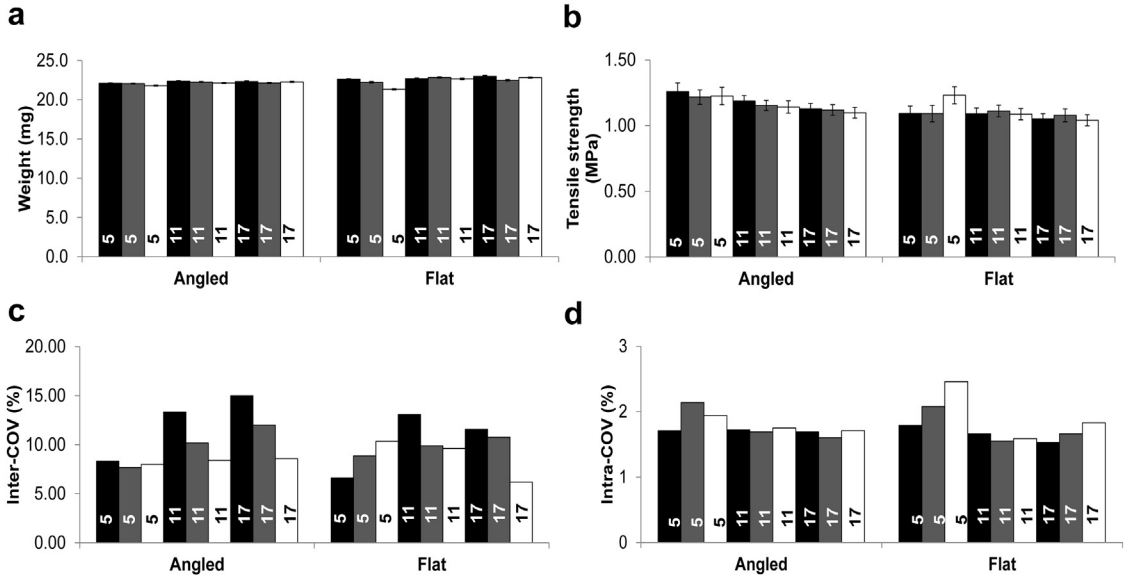


Figure 3. Effects of the paddle angle, paddle speed, and turret speed on (a) average weight, (b) tensile strength, (c) inter-COV, and (d) intra-COV for 3 mm mini-tablets. Turret speed: (■) 25 rpm, (▒) 35 rpm, and (□) 45 rpm. The numbers annotated within the bar graphs indicate the paddle speed in rpm. Error bars indicate CIs.

relationship was relatively linear for both 1.8 mm and 3 mm mini-tablets. Variation in compression roller displacement could be used as a surrogate indicator of mini-tablet weight variation across compaction cycles, as shown previously in another study.[28] Moreover, inter-COV had reflected mini-tablet weight variation across all the compaction cycles ($n = 405$) during the run while inter-COV_{Ref} was only indicative of sampled cycles ($n = 25$).

Inter-$COV_{1.8}$ had a greater spread between 10% and 35% compared to 5% and 15% for inter-COV_3, which reflected greater die fill variability in the smaller mini-tablets (Figs. 2c and 3c). Although inter-$COV_{1.8}$ was slightly higher for mini-tablets produced with the flat paddle ($p < 0.05$), inter-COV_3 appeared to be comparable between the two paddle designs ($p > 0.05$). Increase in the paddle speed was associated with increase in inter-COV albeit the effect was only significant between 17 rpm and 5 rpm for 1.8 mm mini-tablets ($p < 0.05$). The turret speed did not significantly influence inter-COV for both mini-tablet diameters ($p > 0.05$).

Effects of Process Parameters on Intracycle Mini-Tablet Weight Variation

Intra-$COV_{1.8}$ ranged between 5% and 8% although intra-COV_3 was <3% (Figs. 2d and 3d), reflecting the greater difficulty in achieving uniform die fill with the narrower die orifice. Intra-$COV_{1.8}$ values were slightly lower with the angled paddle ($p < 0.05$), whereas intra-COV_3 values were comparable between paddle types ($p > 0.05$). No apparent effect of the paddle speed on intra-$COV_{1.8}$ could be observed ($p > 0.05$). In contrast, intra-COV_3 decreased at

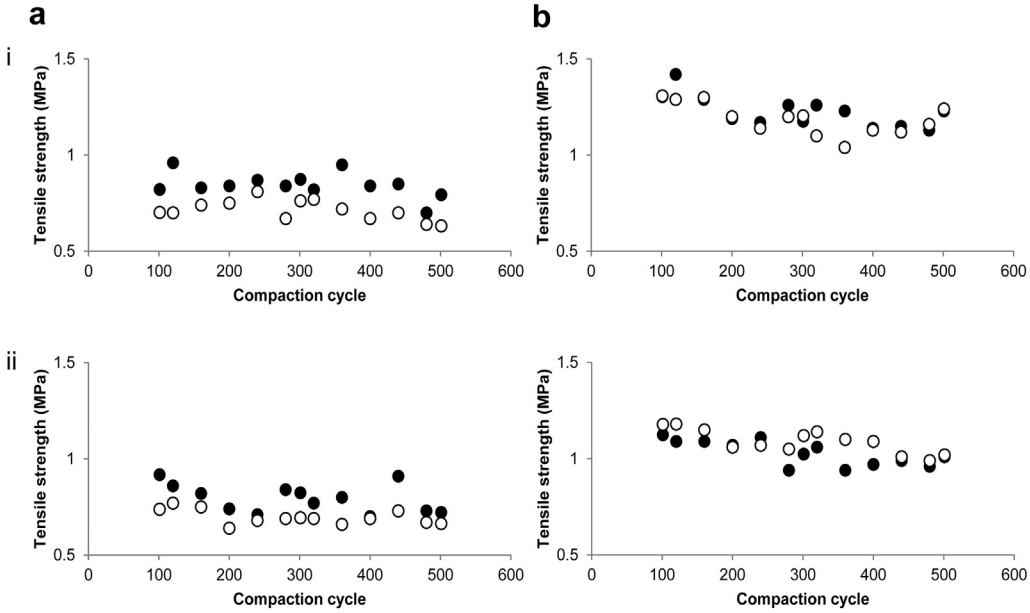


Figure 4. Tensile strength of (a) 1.8 mm and (b) 3 mm mini-tablets produced at (i) 5 rpm and (ii) 17 rpm feed paddle speeds. Paddle profile: (●) flat and (○) angled. Turret speed was 45 rpm for all configurations shown.

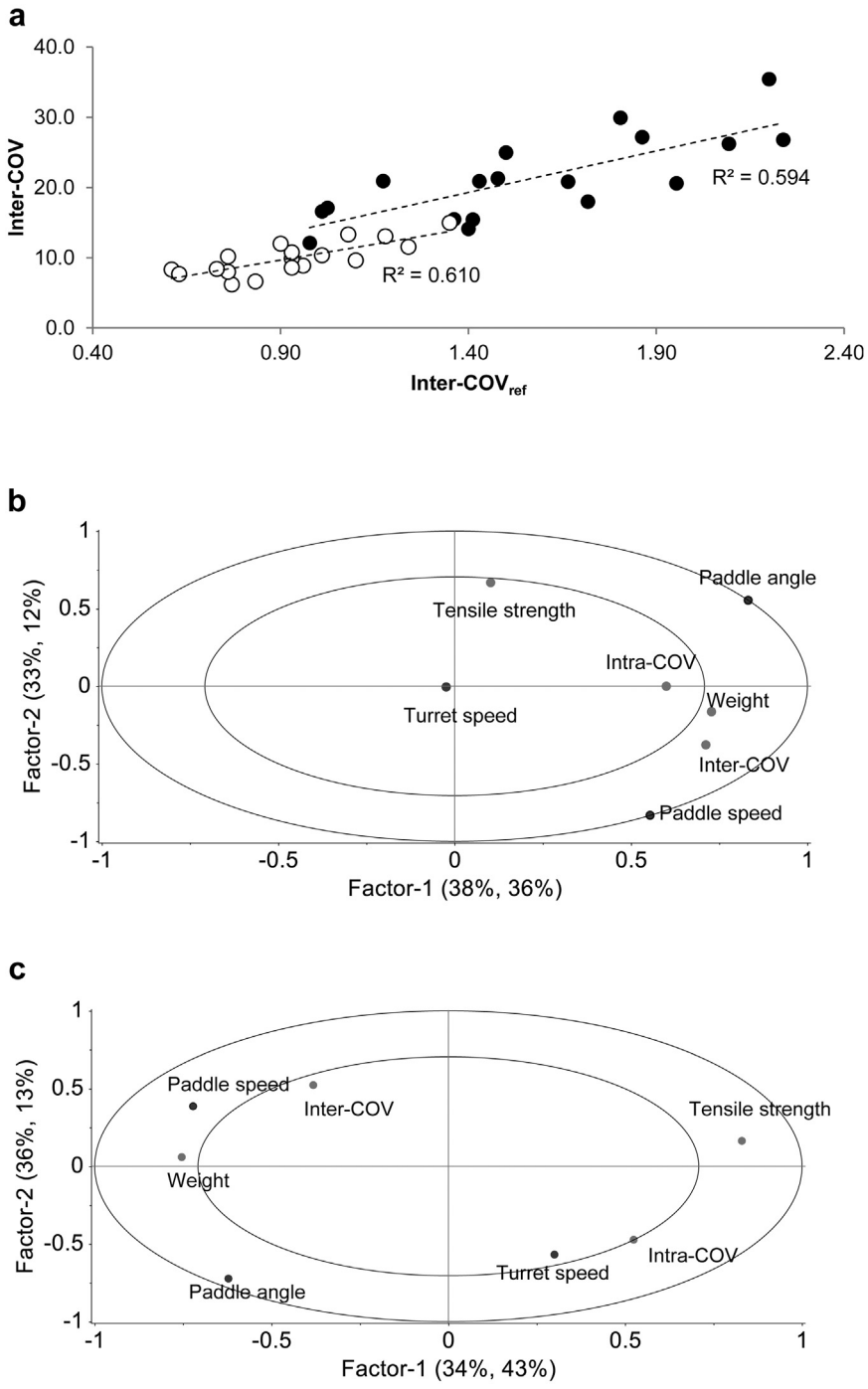


Figure 5. (a) Relationship between inter-COV$_{Ref}$ and (●) inter-COV$_{1.8}$ and (○) inter-COV$_3$. Correlation loadings plots for (b) 1.8 mm and (c) 3 mm mini-tablets.

higher paddle speeds ($p < 0.05$). No clear trend of the turret speed on intra-COV$_{1.8}$ and intra-COV$_3$ could be observed.

Relative Importance of Process Parameters on Mini-Tablet Properties

Partial least squares regression modeling was carried out for each mini-tablet diameter to consider the data in its entirety and elucidate relative contributions of process parameters on mini-tablet properties. The paddle angle, paddle speed, and turret speed were predictor variables, whereas mini-tablet weight, tensile strength, intra-COV, and inter-COV were response variables. Figure 5b and c show the correlation loadings plots. R^2 and root mean square error values of the regression models are shown in Table 3.

For 1.8 mm mini-tablets, 71% of variation in the predictor variables explained 48% of variation in the response variables (Fig. 5b). The paddle speed and paddle angle were significant predictor variables in the model. The turret speed had the least impact on mini-tablet properties because it was located within the inner ellipse and accounted for less than 50% variation in the data set. Variation in weight and inter-COV$_{1.8}$ were better explained by the model as they were located outside the inner ellipse, whereas

Table 3
R^2 and RMSE Values of PLSR Models

Tablet diameter (mm)	Response Variable	R^2	RMSE
1.8	Weight	0.61	0.15 mg
	Tensile strength	0.48	0.17 MPa
	Inter-$COV_{1.8}$	0.67	0.14%
	Intra-$COV_{1.8}$	0.41	0.25%
3	Weight	0.57	0.15 mg
	Tensile strength	0.72	0.16 MPa
	Inter-COV_3	0.42	0.20%
	Intra-COV_3	0.50	0.17%

tensile strength and intra-$COV_{1.8}$ accounted for less than 50% of the explained variation.

Weighted regression coefficients of the paddle angle, paddle speed, and turret speed on 1.8 mm mini-tablet weight and inter-$COV_{1.8}$ are shown in Figures 6a and 6b. Variables with a large regression coefficient typically have important effect on the response. A positive regression coefficient indicates a positive link with the response and the reverse is true for a negative coefficient. Predictors with small coefficients are usually negligible in the regression model. Both the paddle angle and paddle speed were positively associated with mini-tablet weight and inter-$COV_{1.8}$ with a slightly greater influence by the paddle speed.

For 3 mm mini-tablets, 70% of variation in the predictor variables accounted for 56% of variation in the response variables (Fig. 5c). Similar to that observed for 1.8 mm mini-tablets, the turret speed had the least effect on mini-tablet attributes as compared to the paddle angle or speed. Variation in mini-tablet weight and tensile strength were better explained by the model compared to intra-COV_3 and inter-COV_3. Weighted regression coefficients of the paddle angle, paddle speed, and turret speed on 3 mm mini-tablet weight and tensile strength are shown in Figures 6c and 6d. The paddle angle and paddle speed were positively associated with the mini-tablet weight but negatively associated with tensile strength.

Low R^2 values of the regression models indicate that the predictor variables only accounted for part of the variability seen in the response variables. It should be noted that besides process variables investigated in this study, other factors could have also contributed to the mini-tablet properties such as possible changes in powder flow properties in the feed frame during production due to overlubrication and particle attrition.[1,2] Nonetheless, results from the regression analyses identified paddle angle and paddle speed as main process factors influencing mini-tablet properties, which corroborated with observations from Figures 2 and 3. The multivariate analyses also considered possible interactions and correlations among the variables, revealing which of the mini-tablet properties were better accounted for by the process factors as a whole.

Relationship Between N_p and Mini-Tablet Die Fill Performance

N_p represented the number of paddles that passed across the die as it traveled under the feed frame in the die fill region. It was of interest to investigate if N_p could be related to mini-tablet die fill performance. Figure 7 shows the relationship between the N_p and mini-tablet weight, intra-COV, and inter-COV for both mini-tablet diameters.

Average weights of both 1.8 mm and 3 mm mini-tablets increased nonlinearly with N_p via a power function with a plateau effect seen at around N_p of 4.4 regardless of paddle profile (Figs. 7a(i) and 7b(i)). Although no apparent trend was found between intra-$COV_{1.8}$ and N_p (Fig. 7a(ii)), intra-COV_3 decreased with increasing N_p via a power law relationship (Fig. 7b(ii)). N_p was also positively associated with inter-$COV_{1.8}$ for both paddle designs (Fig. 7a(iii)) and inter-COV_3 for the angled paddle (Fig. 7b(iii)).

Discussion

Paddle Angle and Mini-Tablet Properties

The flat paddle was expected to produce lower die fill densities because it simply sweeps feed material horizontally across the die, causing fluidization while exerting little downward force. In contrast, the angled paddle could potentially achieve a greater forced feeding effect as some of the force is directed downward,

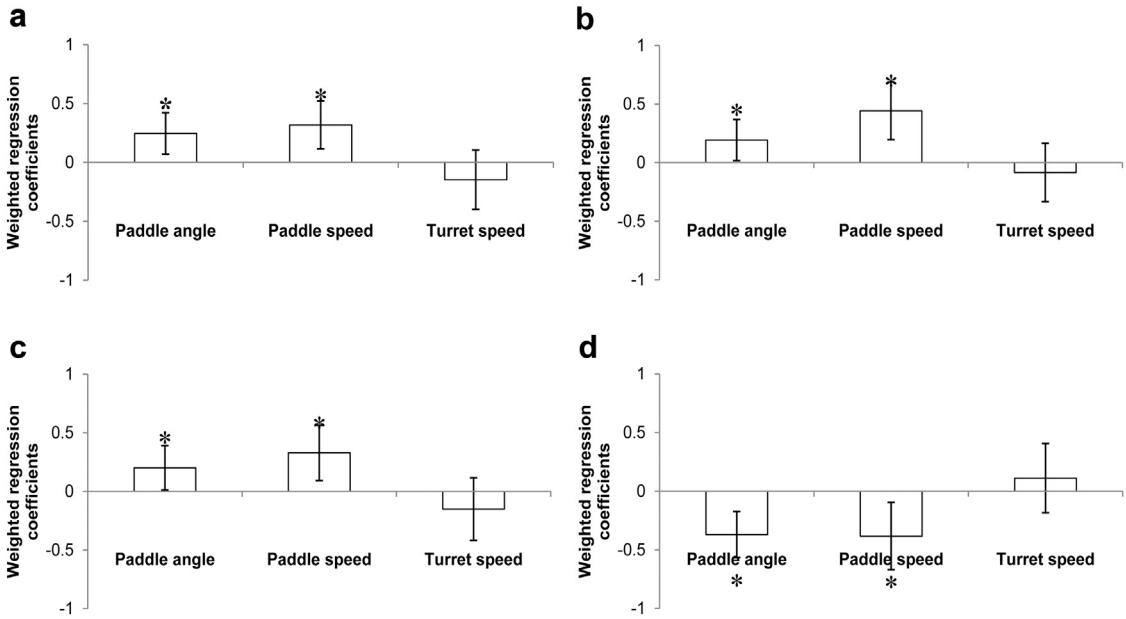


Figure 6. Weighted regression coefficients of the paddle angle, paddle speed, and turret speed on (a) 1.8 mm mini-tablet weight, (b) inter-$COV_{1.8}$, (c) 3 mm mini-tablet weight, and (d) 3 mm mini-tablet tensile strength. Error bars indicate CIs. * indicates factors with significant effects.

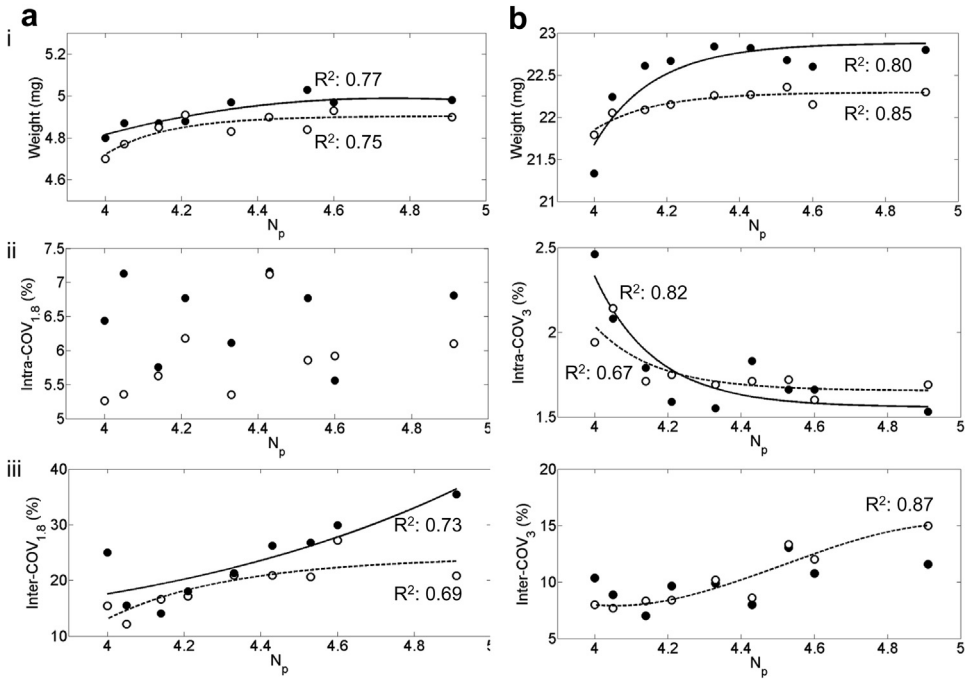


Figure 7. Relationship between N_p and (i) mini-tablet weight, (ii) intra-COV, and (iii) inter-COV for (a) 1.8 mm and (b) 3 mm mini-tablets. Paddle profile: (●) flat and (○) angled. Solid and dashed lines represent fitted trend lines for the flat and angled paddles, respectively.

toward the die orifice as it passes under the feed frame. On the contrary, multivariate analyses revealed that mini-tablets produced with the flat paddle were heavier (Figs. 6a and 6c). A possible explanation for this could be powder compression inside the feed frame by the angled paddle. Forced feeding only occurred when the die moved underneath the paddles in the die fill region. In the brief moments where the die was not present and powder was not actively fed into the die, the angled paddles still continued to rotate and exert downward force on the powder, compressing it and thus impeding some free flowing behavior of powder. The gap between the top of the die table and the paddles also created a region of high-shear conditions,[7] which accentuated the compression effect. Subsequently, active forced feeding into the die had to contend with a more cohesive powder mass. In contrast, the force exerted by the flat paddle on the particles was more tangential to the paddle motion, creating greater fluidization and maintaining powder aeration to facilitate die fill.

Another possible reason for the reduction in die fill density with the angled paddle could be its greater surface area of contact with the granules. Each flat paddle had an estimated contact area of approximately 403 mm^2, whereas that of the angled paddle was 453 mm^2. Force from the angled paddle was thus distributed across a larger number of particles, reducing the overall forced feeding effect despite more of the force being directed downward.

The flat paddle had also resulted in higher inter-$COV_{1.8}$ compared to the angled paddle (Fig. 6b). The compression effect by the angled paddle, while possibly reducing die fill as discussed previously, could also reduce intercycle die fill variation by maintaining a more consistent packing of the granules directly above the die. The effect was only significant for inter-$COV_{1.8}$ possibly due to the greater sensitivity to variation in particle packing with the shallower fill depth.

For 3 mm mini-tablets, the fluidization effect with the flat paddle could have also increased granule overlubrication and reduced mini-tablet tensile strength (Fig. 6d). MgSt is sensitive to mixing intensity. Overmixing can result in formation of lubricant films over the substrate particles and poor interparticle bond formation during compaction, leading to poorer mechanical strength of compacts.[29,30]

Paddle Speed and Mini-Tablet Properties

As the paddle speed increased, feed material was delivered to the die fill region at a higher rate, which contributed to higher die fill densities (Figs. 6a and 6c). The downside of this effect was increased mixing of the lubricated powder within the feed frame, contributing to overlubrication. The paddle speed was thus negatively associated with tensile strength of 3 mm mini-tablets (Figs. 3b and 6d).

The greater degree of fluidization at a higher paddle speed also reduced the time available for particle rearrangement before die entry. Higher intercycle mini-tablet weight variability was seen, especially for 1.8 mm mini-tablets, where the shallower fill depth setting made it more sensitive to the fluidization effect (Fig. 6b).

Turret Speed and Mini-Tablet Properties

An increase in the turret speed reduces the time for die fill (time that the die stays within the die fill region) and creates a stronger suction to draw powder into the die as the lower punch is retracted more quickly. These opposing effects could have masked the effect of the turret speed on mini-tablet weight and weight variation especially because the volume of fill is relatively small for both mini-tablet diameters.

The turret speed also plays an important role in tablet mechanical strength as it affects compression kinetics; an increase in the turret speed narrows the compression profile with simultaneous increase in the strain rate and decrease in the dwell time.[31] The lack of association between the turret speed and mini-tablet tensile strength could suggest that the tablet formulation was not sensitive to the changes in the dwell time or strain rate for the different process configurations used in this study.

Number of Paddle Passes Required for Optimal Die Fill Performance

N_p is a dimensionless parameter that considers both the paddle and turret speeds to determine the relative amount of feed material that the die is exposed to within the die fill region. It is analogous to the number of feed shoe passes across the die orifice in a typical shoe-die model or eccentric, single-station tablet presses. A similar concept of N_p has been discussed previously,[7] but currently no attempt has been made to correlate N_p with mini-tablet die fill performance.

Higher die fill densities were achieved with increased N_p simply as a result of the die being exposed to more feed material albeit this effect was limited to a certain extent (Figs. 7a(i) and 7b(i)). For 3 mm mini-tablets, the steeper gradient of the plot from the flat paddle (Fig. 7b(i)) suggested that the die fill with the flat paddle benefited more from increase in N_p. In contrast, the difference between paddle profiles was more muted for 1.8 mm mini-tablets (Fig. 7a(i)). The lower volume fill requirement for 1.8 mm mini-tablets (one-fifth that of 3 mm mini-tablets) might have reduced interactions between N_p and paddle profile.

Reduction in intra-COV_3 with increasing N_p could be attributed to the improvement in die fill density (Fig. 7b(ii)). For 1.8 mm mini-tablets, the narrow die orifices rendered suction fill determinant to the distribution and filling of granules into the multiple orifices in the same die,[28] which might account for the lack of relationship between intra-$COV_{1.8}$ and N_p. For both mini-tablet diameters, inter-COV generally increased with increasing N_p (Figs. 7a(iii) and 7b(iii)). From Equation 6, it is apparent that high N_p values entailed high paddle speeds that contributed to higher inter-COV as discussed previously.

An N_p value of about 4.4 was optimal for both mini-tablet diameters in this study because it maximized die fill density and minimized intracycle and intercycle weight variations. The paddle and turret speeds could be adjusted to attain the desired N_p value. For the process configurations used in this study, the paddle speed should be kept at 9 or 16 rpm for a turret speed of 25 or 45 rpm, respectively, to attain an N_p value of 4.4.

Feed Frame Residence Time and Overlubrication Effect

The paddle profile and paddle speed impacted mini-tablet tensile strength more than the turret speed due to possible effects of overlubrication inside the feed frame. In particular, 3 mm mini-tablets appeared to be more sensitive to the overlubrication effect (Figs. 4b(i) and (ii) and 6d). This was attributed to two possible scenarios, first, higher lubrication level used for 3 mm mini-tablet production could have accentuated the overlubrication effect or second, overlubrication of the tablet feed had reached steady state for 1.8 mm mini-tablets.

Approximately 44 mg feed was used in each compaction cycle during 1.8 mm mini-tablet production compared to 180 mg feed for 3 mm mini-tablets. Tablet feed residence time in the feed frame was thus much longer during production of smaller mini-tablets. Prolonged mixing initiated granule overlubrication which possibly reached the steady state after the press was conditioned. Moreover, 1.8 mm mini-tablets had lower tensile strength than 3 mm mini-tablets despite the lower level of lubricant and same compaction pressure used. As such, the second scenario seemed more plausible.

Tablet tensile strength is a critical quality attribute that determines the tablet's physical integrity to withstand hazards in postcompaction processing, transportation, and handling by the consumer. Undesired decrease in tensile strength could be a potential issue in mini-tablet production and a possible cause is the low rate of feed consumption that increases likelihood of overlubrication in the feed frame. This could be mitigated by increasing the turret speed to allow increase feed utilization. Alternatively, the amount of feed material in the feed frame could be reduced by using a thinner feed frame or feed wheel with a larger central hub diameter.

Conclusion

In this study, the effects of the feed wheel paddle design, paddle speed, and turret speed on mini-tablet physical properties were investigated. Weight variations of mini-tablets within and across compaction cycles were determined using a methodology based on compression roller displacement data and mini-tablet weight. The use of the flat paddle and high paddle speeds resulted in higher die fill densities for both 1.8 mm and 3 mm mini-tablets. This was related to more effective fluidization of feed material in the feed frame which, depending on the mini-tablet diameter, potentially led to other effects such as increased intercycle weight variation and reduced mini-tablet tensile strength. Consequently, the paddle angle and paddle speed can have varied effects. The turret speed, on the other hand, did not impact mini-tablet weight, weight variation, or tensile strength significantly. A declining trend for 3 mm mini-tablets in tensile strength during compaction was identified and attributed to overlubrication in the feed frame. Both the paddle angle and paddle speed affected the extent of overlubrication. It was also shown that the number of paddle passes in the die fill region could be correlated to die fill performance during mini-tablet production. Findings from this study could be useful for appropriate selection of process parameters or the feeding mechanisms to address problems with die fill variation and tensile strength during mini-tablet production.

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